

Evolution of male sexual characters in the Oriental *Drosophila melanogaster* species group

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SUMMARY Understanding the genetic and molecular mechanisms of morphological evolution is one of the greatest challenges in evolutionary biology. Sexually dimorphic traits, which often evolve at a high rate due to their involvement in mate choice and sexual selection, present unique opportunities for investigating changes in development over short evolutionary distances. Phylogenetic analysis is essential to provide a historical framework for comparative studies of development by establishing the order and polarity of morphological changes. In this report, we apply a new molecular

phylogeny to reconstruct the evolution of male sexual characters in a group of species closely related to the model species *Drosophila melanogaster*. These highly variable traits include wing melanin patterns, the sex comb, and the structure of external genitalia and analia. We show that sexually dimorphic characters can diverge very rapidly among closely related species. More surprisingly, we also find a pervasive pattern of independent origin and secondary loss of male sexual traits in different evolutionary lineages.

INTRODUCTION

In recent years, our understanding of morphological evolution has greatly benefited from the application of genetic and molecular approaches to comparative embryology. Developmental changes responsible for specific differences in morphology may be identified by comparing the expression and function of homologous genes in divergent organisms (Carroll et al. 2001). At present, most comparative work is concentrated on high-level taxa, such as orders of insects (Patel 2000), classes of arthropods (Akam 2000), or even protostomes versus deuterostomes (Holland 2000). However, differences between closely related species are also attracting increasing attention (Doebley and Stec 1993; Dorweiler et al. 1993; Stern 1998; Sucena and Stern 2000; Kopp et al. 2000; Delattre and Felix 2001; Parichy and Johnson 2001; Theron et al. 2001; Wittkopp et al. 2002). A fundamental advantage of working with closely related taxa is the possibility not only to describe what the differences are but also to understand how and why these differences arose.

The extent of genetic divergence between any two evolutionary lineages is a function of time since their separation: the longer the time, the more differences accumulate. Thus, the genetic and molecular bases of morphological differences between closely related species should be relatively simple compared with the differences between higher level

taxa. Species that are closely related to genetic and developmental model systems are of particular interest. By working with these species, we can move beyond the descriptive analysis of correlations between gene expression patterns and morphology. The close similarity between the species' genomes and development opens the possibility for testing the functional importance of interspecific differences by genetic analysis and transgenic techniques (e.g., Skaer and Simpson 2000; Sucena and Stern 2000; Parichy and Johnson 2001).

Understanding the genetic basis of species differences is only the first step: it illustrates the consequences, but not the process, of evolution. To go from a snapshot to a moving picture, these differences must be put in a historical context. What is the ancestral condition, and what is derived? How many genetic changes were involved, and what was the order and polarity of these changes? Do morphological similarities between species reflect a common origin, or did similar traits evolve independently in separate lineages? If the traits are convergent, do they share a similar genetic basis? Does the evolution of morphology correlate with changes in life history, behavior, or biogeographic distribution? These are all inherently historical questions, and they can only be answered if both taxon sampling and the analysis of comparative data are guided by explicit phylogenetic models.

In this report, we examine the evolution of several morphological traits in the *Drosophila melanogaster* species

group using a newly reconstructed molecular phylogeny of this lineage (details of phylogeny reconstruction are presented elsewhere [Kopp and True, 2002 in press]). The *melanogaster* species group is one of the largest and most diverse in *Drosophila*, comprising 174 species by the latest count, and is traditionally subdivided into up to 11 “subgroups” based on morphological criteria (Bock and Wheeler 1972; Bock 1980; Lemeunier et al. 1986; Toda 1991; Schwaroch 2000). As its name implies, this species group includes *D. melanogaster*—the organism whose developmental genetics is understood better than in any other metazoan. This fact, together with its morphological diversity, makes the *melanogaster* species group one of the most promising models for understanding the evolution of development. We are particularly interested in the so-called “Oriental” species subgroups, which, based on their morphology and chromosomal structure, are especially closely related to *D. melanogaster* (Ashburner et al. 1984; Lemeunier et al. 1986; Lachaise et al. 1988). However, until now, the evolutionary relationships among these subgroups remained unclear, with individual studies either lacking resolution or yielding mutually contradictory results (Lemeunier et al. 1986; Pelandakis et al. 1991; Inomata et al. 1997; Goto and Kimura 2001).

We are primarily concerned with the evolution of sexually dimorphic characters. In many groups of animals, including *Drosophila*, male sexual characters can differ dramatically between closely related species, suggesting that they evolve at a higher rate than most other morphological traits (Eberhard 1985; Andersson 1994; True et al. 1997; Civetta and Singh 1998; Kopp et al. 2000). This rapid evolution is thought to be driven mainly by sexual selection. Traditional models of sexual selection have emphasized “run-away” coevolution between male sexual traits and female preferences for these traits by a positive feedback mechanism (Fisher 1958; Lande 1981; Kirkpatrick 1982). More recently, antagonistic sexual selection, driven by divergent reproductive interests of males and females, has been proposed as an alternative explanation for the “evolutionary arms race” phenomenon (Holland and Rice 1998; Arnqvist et al. 2000; Gavrillets 2000; Hosken et al. 2001; Panhuis et al. 2001). Despite the differences in the implied nature of selective pressures, both mechanisms predict rapid evolution of sexually dimorphic traits and frequent origin of new male sexual characters, a pattern consistent with empirical observations. This feature of sexually dimorphic characters makes them very useful for exploring changes in development among closely related species, and sometimes even within species.

We address the evolution of several male-specific traits, including wing pigmentation, sex comb structure, and genital morphology, in the Oriental lineage of the *melanogaster* species group. We find that all these traits evolve very rapidly and show little correlation with phylogenetic relation-

ships. Parallel or convergent origin of similar characters in separate clades is pervasive; at the same time, closely related species can have dramatic differences in sexual traits. Understanding these complex patterns of morphological evolution provides the historical framework for investigating its genetic and molecular bases.

MATERIALS AND METHODS

Taxon sampling and phylogenetic framework

We examined representatives of seven Oriental species subgroups: *eugracilis*, *ficuspila*, *melanogaster*, *suzukii*, *takahashii*, *elegans*, and *rhopaloo*. Species belonging to the *ananassae* and *montium* subgroups, which fall outside the Oriental lineage, were used as outgroups. A total of 17 ingroup and 3 outgroup species were analyzed. Species used in this study and their subgroup affiliations are listed in Table 1. For species descriptions and the taxonomy of the *melanogaster* species group, see Bock and Wheeler (1972), Bock (1980), Lemeunier et al. (1986), Toda (1991), and Sultana et al. (1999).

Phylogenetic reconstruction was based on DNA sequences from four nuclear and two mitochondrial loci; up to 4271 bases of total sequence were available for each species. Both separate and combined analyses of the data were performed. Phylogenetic trees were reconstructed using maximum parsimony, maximum likelihood, and Bayesian inference; for each data set, all methods produced

Table 1. Species used in the analysis and their subgroup affiliations

Species subgroup	Species	Abbreviation
<i>melanogaster</i>	<i>melanogaster</i>	mel
	<i>teissieri</i>	tei
	<i>orena</i>	ore
<i>takahashii</i>	<i>takahashii</i>	tak
	<i>lutescens</i>	lut
	<i>trilutea</i>	tri
	<i>pseudotakahashii</i>	pst
	<i>prostipennis</i>	pro
<i>suzukii</i>	<i>suzukii</i>	suz
	<i>biarmipes</i> ¹	bia
	<i>mimetica</i>	mim
	<i>lucipennis</i>	luc
<i>elegans</i>	<i>elegans</i>	ele
	<i>gunungcola</i>	gun
<i>rhopaloo</i>	<i>fuyamai</i>	fuy
<i>ficuspila</i>	<i>ficuspila</i>	fic
<i>eugracilis</i>	<i>eugracilis</i>	eug
<i>ananassae</i>	<i>ananassae</i>	ana
	<i>bipunctata</i>	bip
<i>montium</i>	<i>kikkawai</i>	kik

Abbreviations are those used in Figs. 1 and 2. *D. suzukii*, strain HJ, was obtained from Dr. Y. Fuyama (Tokyo Metropolitan University). *D. gunungcola* and *D. trilutea* were obtained from Dr. M. Kimura (Hokkaido University). All other species were obtained from the US National *Drosophila* Species Stock Center.

¹*D. biarmipes* is synonymous with *D. rajasekari* (Bock and Wheeler 1972; Bock 1980).

identical or compatible results. Most nodes of the tree shown in Fig. 1 are strongly supported by high bootstrap and posterior probability values, by statistical tests, and by congruence among independent data sets. Details of phylogenetic reconstruction and its implications for taxonomy are presented elsewhere (Kopp and True, 2002 in press). Here, we only remark that the *suzukii* subgroup (Table 1) is undoubtedly polyphyletic.

Character optimization and concentrated changes test were performed in MacClade v4.03 (Maddison and Maddison 2001). All characters were treated as unordered, and all changes were considered reversible and equally likely.

Preparation and analysis of cuticular structures

Adult wings and midlegs from flies at least 3 days posteclosion were dehydrated in *n*-propanol and mounted in Euparal (Carolina Biological Supplies, Burlington, NC). Male terminalia were dissected and mounted in Hoyer's medium (Ashburner 1989). For wings, midlegs, and terminalia, light imaging was performed under bright field on a Zeiss Axiophot microscope. For scanning electron microscopy of the sex combs, adult forelegs were dehydrated in ethanol, air dried, coated with gold, and imaged on a Hitachi S570 SEM microscope. Adobe Photoshop software was used for digital image processing.

Immunohistochemical staining of pharate adult wings

Pharate adult wings (80–100 h after pupariation) were excised in phosphate-buffered saline (PBS) solution, flattened in fixative containing 4% formaldehyde in PBST (PBS + 0.1% Triton X-100) under a coverslip for 5–10 min, and placed in a glass scintillation vial containing the same fixative for 30–60 min on ice. Halfway through fixation, specimens were sonicated for 30–60 sec in a Branson 200 jewelry cleaner (Branson Ultrasonics Corp., Danbury, CT). All subsequent steps until mounting were performed at 4°C. Samples were blocked in PBST+3% bovine serum albumin (BSA) for at least 1 h, incubated with rat anti-Yellow antibody (Wittkopp et al. 2002) at 1:200 dilution in PBST+BSA overnight, washed four times for 15 min each in PBST, and incubated with biotinylated anti-rat secondary antibody (Jackson Laboratories, Bar Harbor, ME) at 1:200 dilution for at least 4 h. Samples were then washed as above, incubated in FITC-conjugated streptavidin (Jackson Laboratories, Bar Harbor, ME) at 1:200 dilution for 2 h, washed as above, mounted in Vectashield (Vector Labs, Burlingame, CA), and imaged on a Bio-rad MRC 1024 confocal microscope.

RESULTS AND DISCUSSION

Taxon sampling

The *melanogaster* species group is one of the largest and most diverse species groups in the genus *Drosophila*, comprising as many as 174 species according to the latest review of taxonomic literature (Schawaroch 2000). It is traditionally subdivided into three main lineages—the *ananassae* species subgroup, the *montium* subgroup, and the so-called Oriental lineage consisting of seven closely related species subgroups (*melanogaster*, *suzukii*, *takahashii*, *elegans*, *rhopaloea*, *ficus-*

phila, and *eugracilis*) (Bock and Wheeler 1972; Bock 1980; Ashburner et al. 1984; Lemeunier et al. 1986; Toda 1991). The position of three other subgroups (*denticulata*, *flavohirta*, and *longissima*) has not been established, and a number of species remain unassigned (Bock 1980; Schawaroch 2000).

In this report we limit our analysis to the Oriental lineage. This lineage appears to be monophyletic based on chromosome structure and molecular evidence (Ashburner et al. 1984; Lemeunier et al. 1986; Pelandakis et al. 1991; Inomata et al. 1997; Goto and Kimura 2001; Schawaroch 2002 in press), and yet it shows remarkable morphological diversity (Bock and Wheeler 1972; Bock 1980; Toda 1991). Together, the seven Oriental subgroups comprise 55 species, according to the latest count (Table 2) (Bock and Wheeler 1972; Bock 1980; Toda 1991; Sultana et al. 1999; Schawaroch 2000). Unfortunately, most of these species are only known from museum specimens and have been assigned to subgroups based on phenetic criteria. This classification is of little use in reconstructing morphological evolution, because it sheds no light on the relationships either within or among subgroups. Moreover, recent molecular phylogenies reveal that the *suzukii* subgroup is polyphyletic (Schawaroch 2002; Kopp and True, 2002 in press), and the same cannot be ruled out for some of the other subgroups, as well. Therefore, our strategy has been to use a molecular phylogeny (Kopp and True, 2002 in press) as an independent framework for the reconstruction of morphological changes. This necessarily limits our analysis to only those species for which molecular data could be obtained. The resulting taxon sample comprises 17 species, not including additional members of the well-studied, monophyletic *melanogaster* subgroup. Although this is the largest number of Oriental species studied to date, it is still only one third of the total number of species in this lineage. However, this sample is roughly representative of the Oriental lineage as a whole in terms of both taxonomic and morphological diversity (Table 2).

Wing spots and mating displays

Melanization of the anterior-distal wing in males has been described in several members of the *takahashii*, *suzukii*, *elegans*, and *rhopaloea* species subgroups (Bock 1980; Toda 1991). This is clearly a recently derived trait: no wing melanization is seen in the *ananassae* and *montium* subgroups, which are the nearest relatives of the Oriental lineage, or in any of the closest relatives of the *melanogaster* species group (subgenus *Sophophora*). Although melanic spots always appear in roughly the same area of the wing, the shape and intensity of the spot differ considerably among species (Fig. 1). The simplest pattern is seen in *D. eugracilis*, in which both sexes show a slight variable darkening of the anterior of the wing, mostly in the L1-2 and L2-3 intervein areas (Fig. 1). This pattern is somewhat stronger in males than

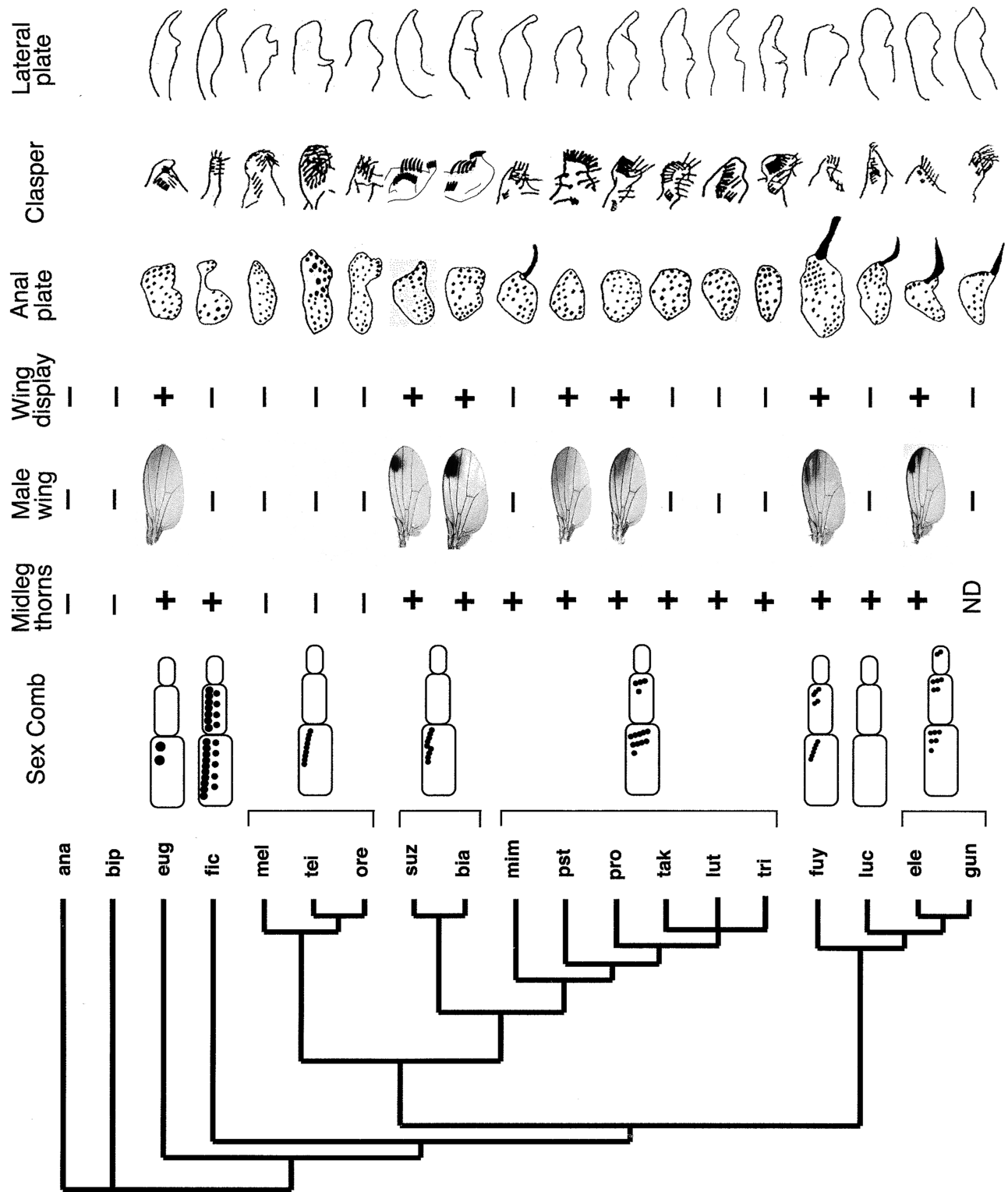


Fig. 1. Evolution of male sexual characters in the Oriental lineage. Details of phylogeny reconstruction are presented elsewhere (Kopp and True, unpublished data). For the sex comb, the first three tarsal segments of the male foreleg are shown schematically, with the distal segments to the left. “-” in the male wing column indicates that the wing is clear. For the anal plates, lateral plates, and claspers, lateral views are shown. The drawings of the male terminalia were made by direct tracing of photographs or reproduced from original sources (Bock and Wheeler 1972; Sultana et al. 1999), with permission.

Table 2. Taxon sampling

Species subgroup	Total number of described species	Number of species with wing spots	Sex comb morphology ¹	Number of sampled species
<i>melanogaster</i>	8	0	R	3
<i>suzukii</i>	17	4	R,T,A	4
<i>takahashii</i>	13	5	T	5
<i>elegans</i>	5	4	T	2
<i>rhopaloea</i>	5	3	R,T,A	1
<i>ficuspshila</i>	6	0	R	1
<i>eugracilis</i>	1	0	T	1

The total number of described species is from Schawaroch (2000).

¹R, rotated sex comb; T, transverse sex comb; A, sex comb absent.

in females. A more intense but diffuse melanization of the anterior-distal wing, covering approximately one fourth of the wing area, is seen in males of *D. prostipennis* and *D. pseudotakahashii* (Fig. 1). Sexual dimorphism is more pronounced in these species, with females showing only a barely perceptible darkening. In *D. elegans* and *D. fuyamai*, melanization occurs in males only and follows the distal L-2 and L-3 veins and the wing margin (Fig. 1). Finally, in *D. suzukii* and *D. biarmipes*, the melanin pattern takes the form of an intense spot centered on the distal L-2 vein (Fig. 1). This pattern is also limited to males, although very slight melanization is occasionally seen in females of *D. biarmipes*.

Phylogenetic reconstruction reveals that wing melanization is not limited to a single evolutionary lineage but is present sporadically in several distinct clades (Figs. 1 and 2A). The evolution of male wing spots clearly involved multiple independent gains or multiple independent losses. For instance, although *D. elegans* and *D. fuyamai* show very similar melanization patterns, *D. gunungcola* and *D. lucipennis* (both of which are more closely related to *D. elegans* than is *D. fuyamai*) lack wing spots altogether (Fig. 1). Similarly, melanization is present in *D. prostipennis* and *D. pseudotakahashii*, the basal species of the *takahashii* subgroup, but absent both in the more derived members of that subgroup and in *D. mimetica*, which branches more basally than the pigmented species (Fig. 1).

The most parsimonious reconstruction of wing spot evolution requires six steps—four gains and two losses (Fig. 2A). Because the wing spot is present in only seven taxa, its phylogenetic distribution is not significantly different from random, and its retention and consistency indices are predictably low (RI = CI = 0.17; Fig. 2A). However, given the basal phylogenetic position of *D. eugracilis*, it is possible that the slight weakly dimorphic melanization of the anterior wing seen in that species represents the ancestral state for the entire Oriental lineage. Subsequently, this character could become exaggerated and more sexually dimorphic in some lineages (*D. elegans* + *D. fuyamai* and *D. suzukii* + *D. biarmipes*), whereas other clades would have lost it altogether

(*melanogaster* and *ficuspshila* subgroups). Assuming this scenario, a single gain followed by six independent losses and by the intensification of the pattern in two separate lineages is necessary to account for the phylogenetic distribution of wing spots (Figs. 1 and 2A).

The most important conclusion in this regard is that both loss and gain (or intensification) of wing melanin patterns have occurred independently in multiple evolutionary lineages. This offers an attractive opportunity for investigating the genetic and molecular bases of parallel morphological changes. The development of wing melanin patterns occurs in two distinct steps. Coordinated expression of several enzymes in restricted spatial domains in the pupal wing sets up a prepattern, which is subsequently elaborated in a vein-dependent manner during early adult life (True et al. 1999; Wittkopp et al. 2002). A key element of the prepattern is the extracellular protein Yellow, which is required for dark melanization and is expressed in areas of the epidermis that will secrete melanized cuticle (Walter et al. 1991; Wittkopp et al. 2002). In particular, Yellow expression is upregulated in the male pupal wing in a pattern that prefigures adult pigmentation in *D. biarmipes* (Wittkopp et al. 2002) (Fig. 3A) and other species that show male-specific wing spots (Fig. 3, C, E, and G). However, little or no Yellow expression is seen in females of the same species (Fig. 3, B, D, F, and H). Interestingly, in species with large spots, such as *D. biarmipes* and *D. prostipennis*, Yellow expression extends from the anterior-distal wing margin internally past the presumptive L-3 vein (Fig. 3, A and C). On the other hand, species with vein-associated patterns, such as *D. elegans* and *D. fuyamai*, have narrower domains of Yellow expression associated with the distal L-2 and L-3 veins (Fig. 3, E and G). These observations suggest that evolutionary changes in the regulation of Yellow have played an important role in the evolution of wing melanin patterns.

The presence of wing spots in the Oriental *melanogaster* group species is associated with a peculiar male courtship behavior, in which the male faces and circles the female's head, turning his dorsal wing surfaces toward her and waving them up to a straight overhead position and back

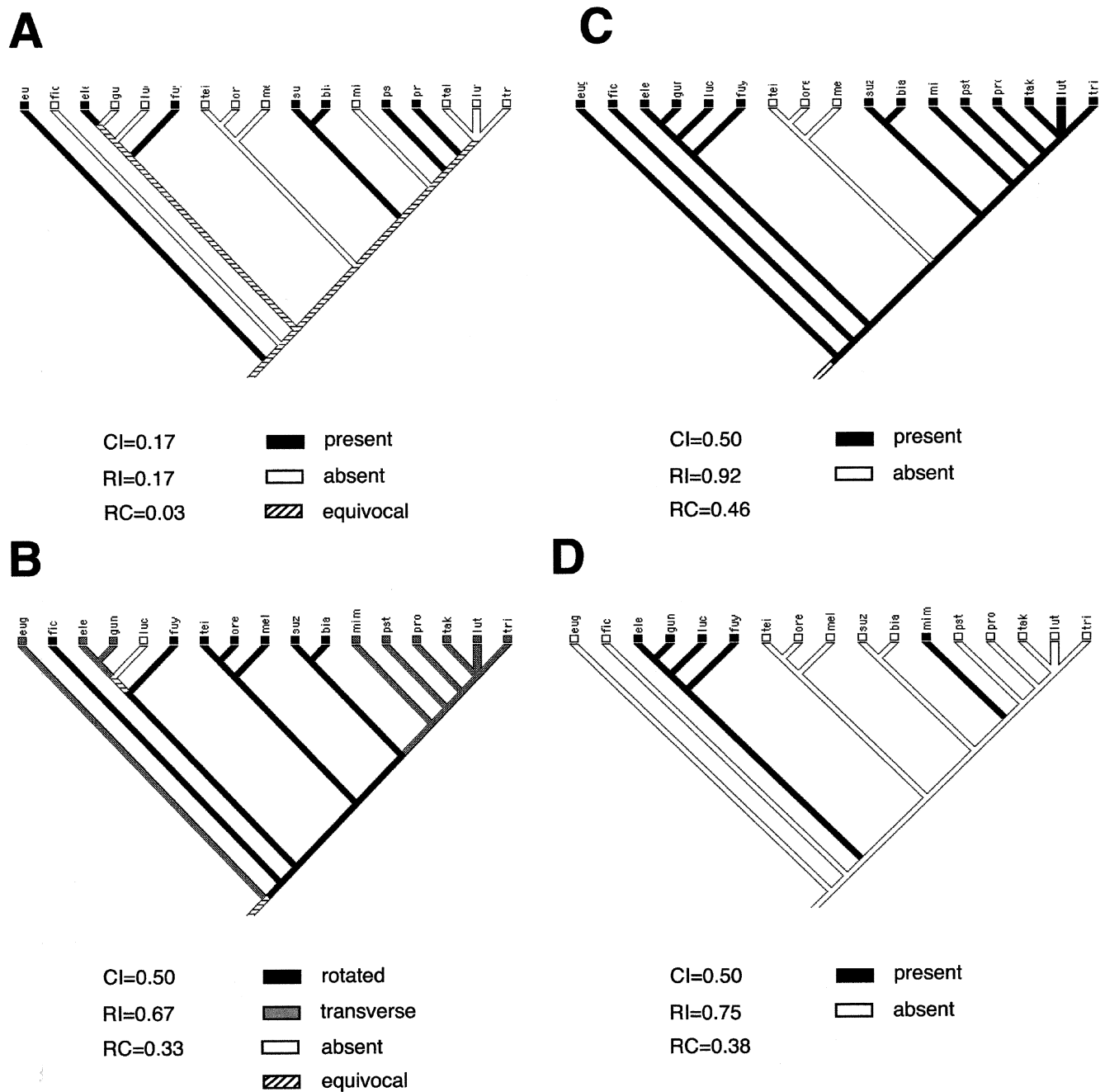


Fig. 2. Most-parsimonious reconstructions of character states. (A) Male wing spot. (B) Sex comb orientation. (C) Midleg thorn bristles. (D) Enlarged anal bristle. CI, consistency index; RI, retention index; RC, rescaled consistency index.

(Fuyama 1979). The association between morphology and behavior is perfect within our taxon sample: species that have male-specific wing spots also possess the courtship display, whereas the unpigmented species do not include this element in their courtship (Fig. 1). Even *D. eugracilis*, which has only a faint and weakly dimorphic wing pigmentation, performs wing display during courtship. The concentrated

changes test (Maddison 1990) shows a highly significant correlation between the two traits ($P = 0.000012$ under the most parsimonious [4 gains/2 losses] scenario; $P = 0.000009$ under the single gain/6 losses scenario). It is tempting to speculate that evolutionary changes in behavior and morphology were part of the same process and were jointly driven by sexual selection.

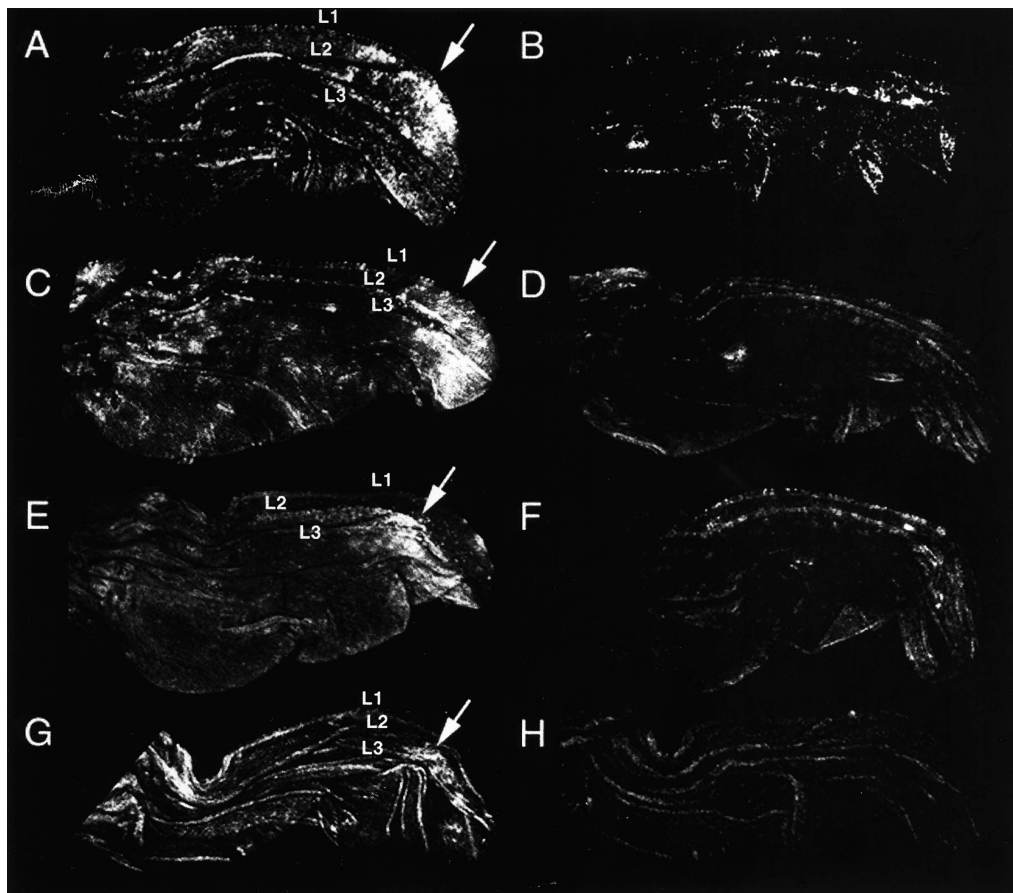


Fig. 3. Yellow protein expression in pharate adult wings prefigures male-specific melanin patterns. Left column: males; right column: females. Arrows indicate the sites of Yellow expression. Positions of the longitudinal veins L-1, L-2, and L-3 in the male wings are also indicated. (A, B) *D. biarmipes*; (C, D) *D. prostipennis*; (E, F) *D. elegans*; (G, H) *D. fuyamai*.

The sex comb

The sex comb of *Drosophila* is a male-specific array of modified bristles that develops on the anterior–ventral surface of the first pair of legs, typically on the first tarsal segment (basitarsus) but sometimes also on the more distal segments. This structure is thought to be involved in male–female interactions during mating (Spieth 1952; Cook 1977), suggesting that evolutionary changes in its morphology could be driven by sexual selection (Markow et al. 1996). The sex comb is a recent evolutionary innovation: it is present only in the monophyletic clade consisting of the *melanogaster* and *obscura* species groups (and, apparently independently, in the genus *Lordiphosa*; Hu and Toda 2001), but it is absent in other *Drosophila* species. The sex comb develops from the tightly packed transverse bristle rows that are present in both sexes and in all species of *Drosophila* and related genera. Although it is present in virtually all members of the *melanogaster* and *obscura* species groups, its morphology shows remarkable diversity (Figs. 1 and 4) (Lakovaara and Saura 1982; Lemeunier et al. 1986). Variable traits include

the number of bristles that comprise the sex comb, the presence of sex combs on the second and third tarsal segments, and the degree of bristle modification. In some species the modified bristles (teeth) remain in place, and the sex comb has a transverse orientation with teeth pointed distally (Figs. 1 and 4). In others, including *D. melanogaster*, the distal part of each tarsal segment that carries a sex comb undergoes a 90-degree rotation in the male, resulting in an oblique orientation of the sex comb with teeth pointed in the anterior direction (Figs. 1 and 4) (see Hannah-Alava 1958; Tokunaga 1962 for descriptions of sex comb structure and development).

We find that sex comb morphology evolves more slowly than wing pigmentation and tends to be similar within each of the major Oriental clades (Figs. 1 and 2B). In *D. eugracilis*, the sex comb consists of two greatly enlarged, straight, distally oriented bristles on the distal basitarsus (Figs. 1 and 4A). In *D. ficusphila*, the sex comb is fully rotated and extends along the entire length of the first and second tarsal segments (Figs. 1 and 4B). The curved and greatly enlarged

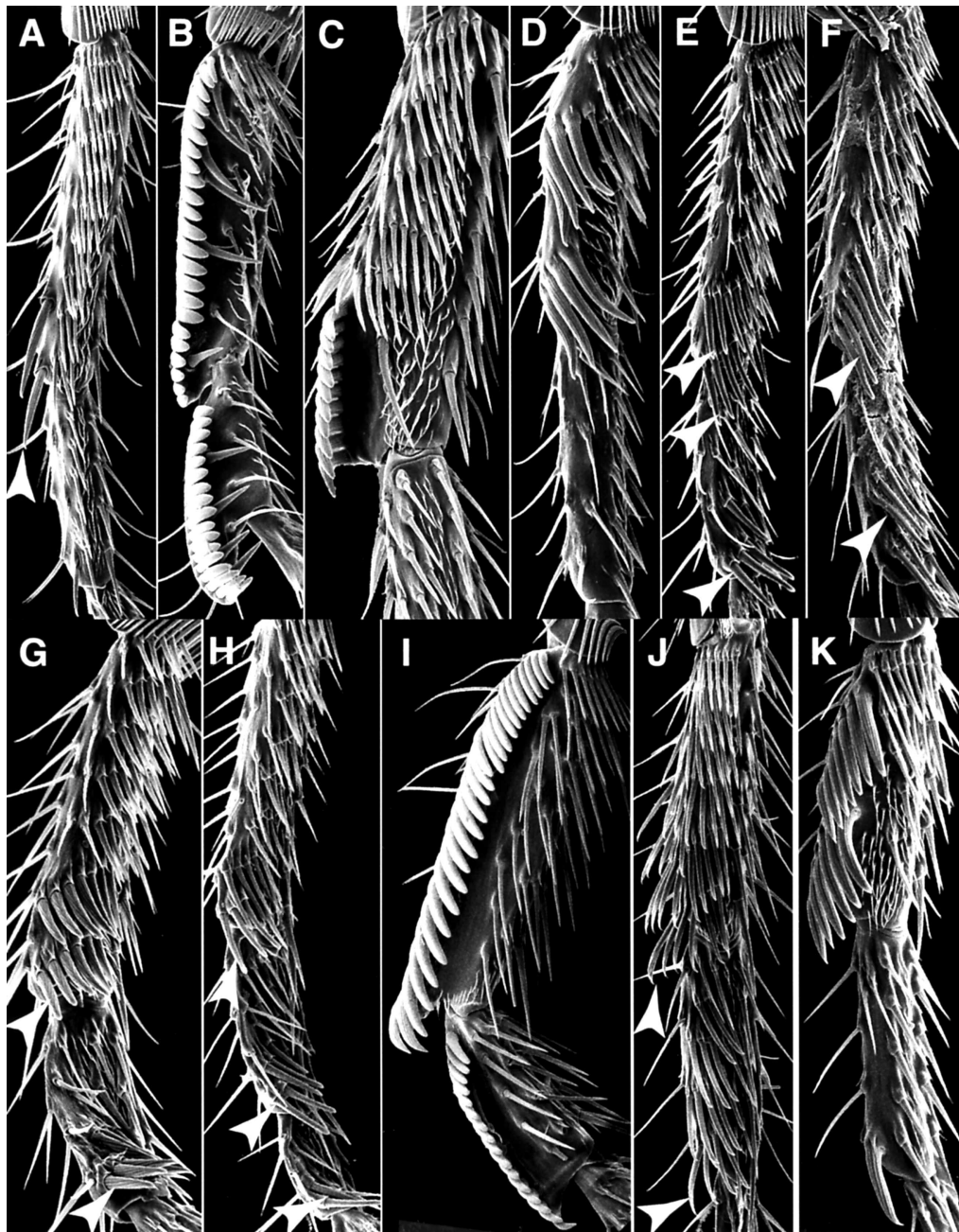


Fig. 4. Diversity of sex comb morphology in the *melanogaster* species group. Distal is down, and anterior is to the left in all panels. (A) The sex comb of *D. eugracilis* consists of two enlarged straight bristles on the distal basitarsus (arrowhead). (B) In *D. ficusphila*, sex comb teeth are greatly enlarged and curved ventrally and arranged in a tightly packed anterior and a sparse posterior rows. (C) *D. melanogaster*. (D) *D. biarmipes*. (E) In *D. takahashii*, the sex comb consists of weakly modified bristles arranged in transverse rows on the distal first and second tarsal segments (arrowheads). (F) In *D. fuyamai*, the sex comb is oblique on the first tarsal segment and transverse on the second (arrowheads). (G) The sex comb of *D. mimetica* consists of thickened bristles in transverse rows on the distal first and second tarsal segments (arrowheads). (H) In *D. elegans*, weakly modified sex comb teeth are present on the first, second, and third tarsal segments (arrowheads). (I) The sex comb of *D. kikkawai* (*montium* subgroup) is similar to that of *D. ficusphila* (B), except that bristles are arranged in a single row. (J) The transverse sex comb of *D. ananassae* (*ananassae* subgroup) consists of very slightly modified teeth, which are only distinguishable from regular bristles by their curved tips (arrowheads). (K) In *D. bipectinata* (*ananassae* subgroup), the oblique sex comb is similar to that of *D. biarmipes* and *D. suzukii* (D). A single tooth is also present on the second tarsal segment.

bristles are packed densely in the main, anterior row, and more loosely in the secondary, posterior row. The sex comb typically contains 28–31 bristles on the first tarsal segment and 21–25 bristles on the second; a single sex comb tooth is occasionally seen on the third tarsal segment. In the *melanogaster* subgroup, the sex comb consists of 9–14 bristles arranged in a single oblique row on the first tarsal segment (Figs. 1 and 4C). The bristles are also curved ventrally but are much smaller than in *D. ficusphila*. Similar sex combs are present in *D. biarmipes* and *D. suzukii*, with the exception that bristles are arranged in two oblique rows rather than one (Figs. 1 and 4D). A different type of sex comb is seen in *D. mimetica*, the *takahashii* subgroup, and the *D. elegans*–*D. fuyamai* clade. In these species, the sex comb consists of straight weakly modified bristles arranged in several transverse rows on the first and second, and sometimes also the third, tarsal segments (Figs. 1 and 4, E–H). The only deviation from this pattern is found in *D. fuyamai*, in which the sex comb on the first, but not the second, tarsal segment is partially rotated (Fig. 4F). Finally, *D. lucipennis* lacks the sex comb altogether (Fig. 1).

Although sex comb morphology retains some phylogenetic signature, similar sex combs are found in separate clades and, conversely, a single clade can include species with different types of sex combs. For instance, oblique sex combs are present in the *melanogaster* subgroup, as well as in *D. biarmipes* and *D. suzukii*, but not in the closest relatives of the last two species (Fig. 1). Similarly, transverse sex combs are found in the clade consisting of the *takahashii* subgroup and *D. mimetica* and, independently, in the *D. elegans*–*D. fuyamai* clade (Fig. 1). Although this latter type of sex comb appears to be a derived state in the Oriental lineage, it is reminiscent of the morphology seen outside this clade, in the *ananassae* subgroup (Fig. 4J). At the same time, the large oblique sex comb of *D. ficusphila* has a strong similarity to another outgroup, the *montium* species subgroup (Fig. 4I), whereas the sex comb of *D. biarmipes* and *D. suzukii* is similar to the structure seen in *D. bipectinata*, a member of the *ananassae* subgroup (Fig. 4K). Thus, the transition between transverse and oblique sex combs has occurred several times in the *melanogaster* species group and in both directions.

The most parsimonious reconstruction of sex comb orientation in the Oriental lineage requires four steps (e.g., one transition from transverse to rotated, two reversions to transverse, and one complete loss of the sex comb; Fig. 2B) and is associated with moderate consistency (0.50) and retention (0.67) indices (Fig. 2B). However, a simple classification of sex combs into rotated and transverse ignores other important aspects of sex comb structure. The evolution of sex comb morphology can be described more accurately as a trajectory in a three-dimensional morphospace (Fig. 5). The three parameters are the orientation (transverse or rotated),

the number of bristles, and the degree of bristle modification. It is apparent that closely related species can occupy very different positions in the morphospace, whereas species that are close to each other in morphospace can arrive there from distant phylogenetic branches (Fig. 5).

The polyphyletic nature of the *suzukii* subgroup has important implications. This subgroup was thought to include species with oblique sex combs (*D. biarmipes* and *D. suzukii*), transverse sex combs (*D. mimetica*), and with no sex comb whatsoever (*D. lucipennis*) (Bock 1980). We find, however, that although the two species with oblique sex combs do form a monophyletic clade, *D. mimetica* is more closely related to the *takahashii* subgroup (which also has a transverse sex comb) and *D. lucipennis* is closer to the *elegans* subgroup and to *D. fuyamai* (Figs. 1 and 2B).

Midleg thorn bristles

In many Oriental species, several thickened hooked bristles are present exclusively in males on the distal tibia and proximal basitarsus on the second pair of legs (Okada 1963; our observations). These bristles, whose function is unknown, range in morphology from straight and relatively slender, as in the *takahashii* subgroup (Fig. 6A), to bulbous, as in *D. eugracilis* (Fig. 6B). Phylogenetic analysis indicates that these bristles originated at the base of the Oriental clade and were subsequently lost in the *melanogaster* subgroup (Figs. 1 and 2C). Secondary loss has also been reported for another sexually dimorphic trait in *Drosophila*, a male-specific abdominal muscle (the so-called Muscle of Lawrence; Lawrence and Johnston 1984; Gailey et al. 1997). Recent phylogenetic studies demonstrate that loss of male sexual traits is common in a variety of animals; one possible explanation is that female mating preferences (and therefore sexual selection) may often disappear or reverse direction (Wiens 2001).

Male genitalia and analia

Male external genitalia and analia are the most rapidly evolving morphological traits in *Drosophila* and are widely used for taxonomic purposes (Hsu 1949; Okada 1954; Bock and Wheeler 1972). Even sibling species, separated by only a few hundred thousand years and identical in all other respects, can almost always be distinguished on the basis of male genital morphology. Genetic analyses suggest that differences between closely related species in the morphology of male terminalia are controlled by multiple loci (Coyne 1983; Liu et al. 1996; True et al. 1997; Macdonald and Goldstein 1999). However, the identity of these loci remains unknown, and the development of genital structures in general is poorly understood (see Sanchez and Guerrero 2001 for a recent review).

We used a phylogenetic approach to reconstruct the evolution of three highly variable terminal structures. Male genitalia in *Drosophila* develop from the ninth abdominal segment

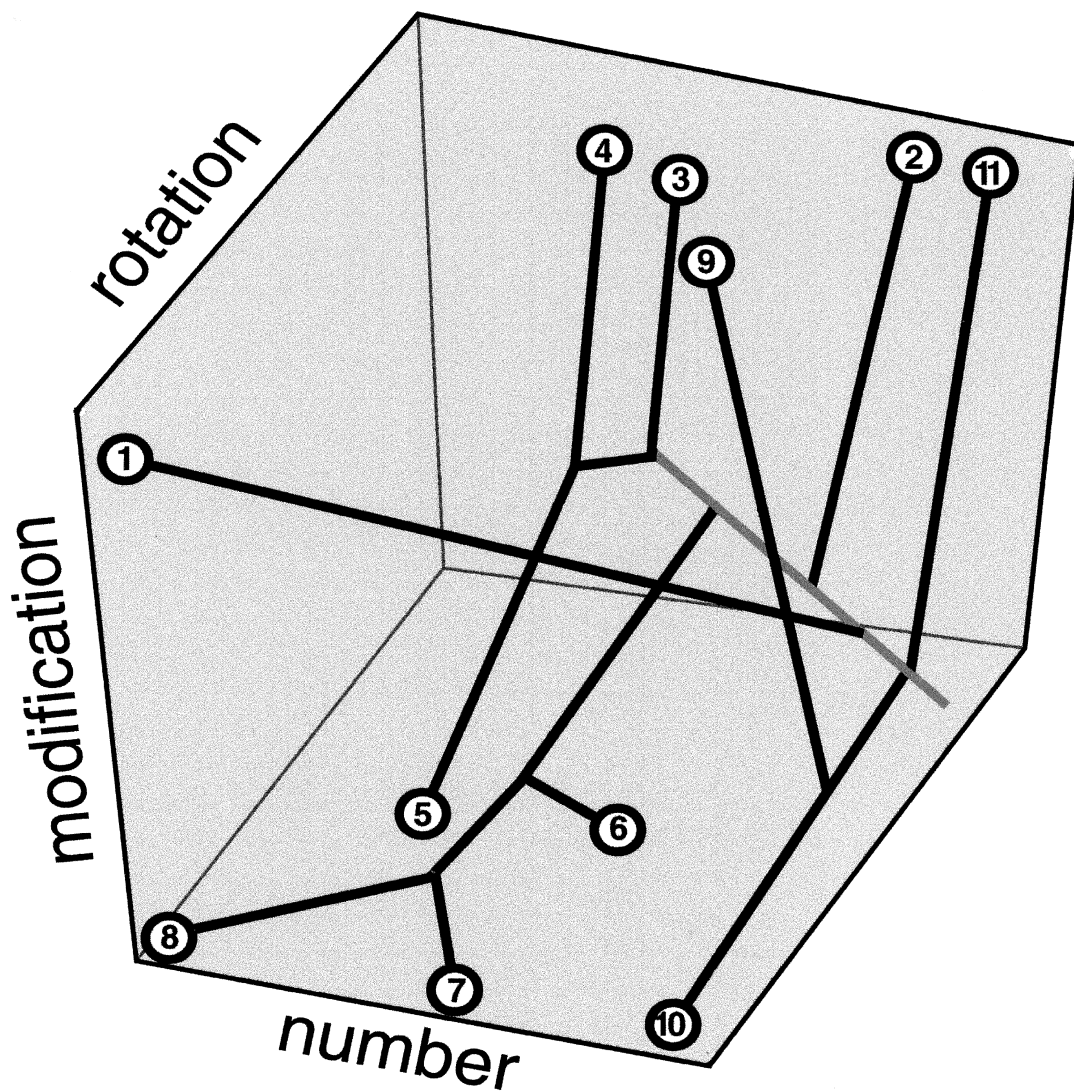


Fig. 5. Evolution of the sex comb in a three-dimensional morphospace. The three parameters are the number of teeth (left, fewer; right, more), the degree of modification (down, weakly modified; up, strongly modified), and the orientation of the sex comb (proximal, transverse; distal, oblique). Thick lines indicate phylogenetic relationships. 1, *D. eugracilis*; 2, *D. ficusphila*; 3, *melanogaster* subgroup; 4, *D. biarmipes* and *D. suzukii*; 5, *takahashii* subgroup and *D. mimetica*; 6, *D. fuyamai*; 7, *elegans* subgroup; 8, *D. lucipennis*; 9, *D. bipectinata*; 10, *D. ananassae*; 11, *D. kikkawai*.

and consist of a bilateral pair of cuticular plates that are fused at the dorsal midline to form the genital arch (Fig. 7A). On each side, the genital arch carries two ventral projections: a large lateral plate (epandrium) and a smaller clasper (surstylus). The Oriental *melanogaster* group species show extensive differences in the shape of the ventral lobe of the lateral plate (the “toe”) and in the presence and shape of the posterior lobe (“epandrial expansion”). The size and shape of the clasper and the number and morphology of clasper bristles (teeth) are also highly variable among species. The external analia develop from the tenth abdominal segment and consist of bilateral anal plates situated dorsally to either side of the

anus (Fig. 7A). Many closely related species differ, often dramatically, in the shape of the anal plate and in the morphology of bristles it carries.

Male terminal characters evolve so rapidly in the *melanogaster* species group, both within and outside the Oriental lineage, that it is often difficult to reconstruct the ancestral states. For instance, the two most basal species, *D. eugracilis* and *D. ficusphila*, are remarkably different in the morphology of anal plates and claspers (Fig. 1). However, both species have a narrow curved ventral lobe of the lateral plate and lack the posterior lobe (Fig. 1), suggesting that these traits may be ancestral in the Oriental lineage.

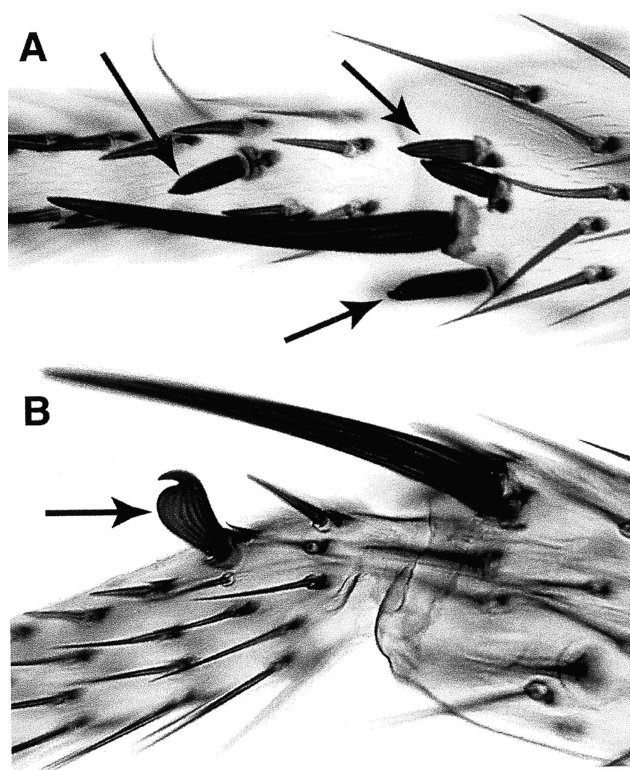


Fig. 6. Midleg thorn bristles. A. In *D. takahashii*, several thickened straight bristles are present on the distal tibia and proximal basitarsus (arrows). B. In *D. eugracilis*, single bulbous hooked bristle is seen on the basitarsus (arrow). The long straight bristle at the distal tip of the tibia is the “apical bristle” present in all *Drosophila* species.

Examination of the phylogenetic distribution of various traits (Fig. 1) highlights the differences that exist among closely related species. We find that male terminal structures contain few synapomorphies (shared derived characters). Most traits are autapomorphic (uniquely derived), and there are also examples of convergent or parallel evolution of similar traits in separate lineages. Below, we consider a few examples of each of the three classes of characters. More detailed descriptions and drawings of male genitalia can be found in taxonomic sources (Bock and Wheeler 1972; Bock 1980; Toda 1991; Sultana et al. 1999).

The most clear synapomorphy is the presence of a single extremely large melanized bristle at the ventral tip of the anal plate in the monophyletic clade consisting of *D. fuyamai*, *D. lucipennis*, and the *elegans* subgroup (*D. elegans* and *D. gunungcola*) (Figs. 1, 2D, and 7, B and C). These species also share a wide rounded ventral lobe of the lateral plate, which may represent a derived character in the Oriental lineage (Fig. 1). Another synapomorphy is found in the *takahashii* subgroup, where all species have a distinctive fan-shaped row of clasper teeth (Figs. 1 and 7G), which distinguishes this subgroup from its closest relatives, *D. mimetica*, *D. su-*

zukii, and *D. biarmipes* (Figs. 1 and 7, E and F). In the *melanogaster* subgroup, members of the *yakuba* species complex (represented by *D. teissieri* and *D. orena* in our study) are characterized by large ventral expansions of the anal plate (cercal claspers) that carry modified tooth-like bristles (Figs. 1 and 7I).

Convergent or parallel change is most prominent in *D. mimetica* (Fig. 2D). In this species, the anal plate carries a single greatly enlarged bristle that is reminiscent of similar bristles in *D. elegans* and its relatives (Figs. 1 and 7, B–D). The phylogenetic position of *D. mimetica* is strongly supported (Kopp and True, 2002 in press) and leaves no doubt about the independent origin of this trait. A similar bristle is also seen in the outgroup species *D. bipectinata* (not shown), suggesting that independent evolution of this structure could be common in the *melanogaster* species group. The narrow constriction between the dorsal and ventral parts of the anal plate in *D. ficusphila* (Fig. 7J) is similar to the morphology seen in *D. orena* (Fig. 7I). Finally, the thickened bristles on the ventral surface of the anal plate in *D. suzukii* (Fig. 7E) and in *D. mauritiana*, a member of the *melanogaster* subgroup (Fig. 7H) (Liu et al. 1996), may also represent a convergent or parallel change, because such bristles are not seen in the closest relatives of either species (Fig. 7F) (True et al. 1997).

Autapomorphic differences between male terminal structures in the *melanogaster* subgroup have been well characterized (Liu et al. 1996; True et al. 1997). The best example is the dramatic difference in the shape of the posterior lobe between two sibling species, *D. simulans* and *D. mauritiana* (Fig. 7, K and L) (True et al. 1997). Other closely related species can also differ strikingly in particular structures. The sister taxa *D. suzukii* and *D. biarmipes*, for example, differ in the number of thorn bristles on the clasper (13–14 versus 4, respectively; Fig. 7, E and F). These differences underscore the ongoing evolution of novel male terminal characters, and it could be informative to examine the extent of intraspecific variation in these traits.

Rapid evolution of external male genitalia is a common phenomenon in insects and other animals and is thought to be due to the intense sexual selection acting on these structures (Eberhard 1985, 1990, 1991; Andersson 1994). Models of sexual selection suggest that antagonistic interactions between males and females lead to an evolutionary “arms race,” whereby males are forced to evolve new characters to overcome females’ resistance to the existing traits (Holland and Rice 1998). Details of male genital morphology could play an important role in the so-called copulatory courtship, which involves the stimulation of the female by the male genitalia during mating (Eberhard 1990, 1991, 2001). However, direct evidence for the role of selection in the evolution of male genitalia remains elusive and must await the identification of genes responsible for the morphological changes.

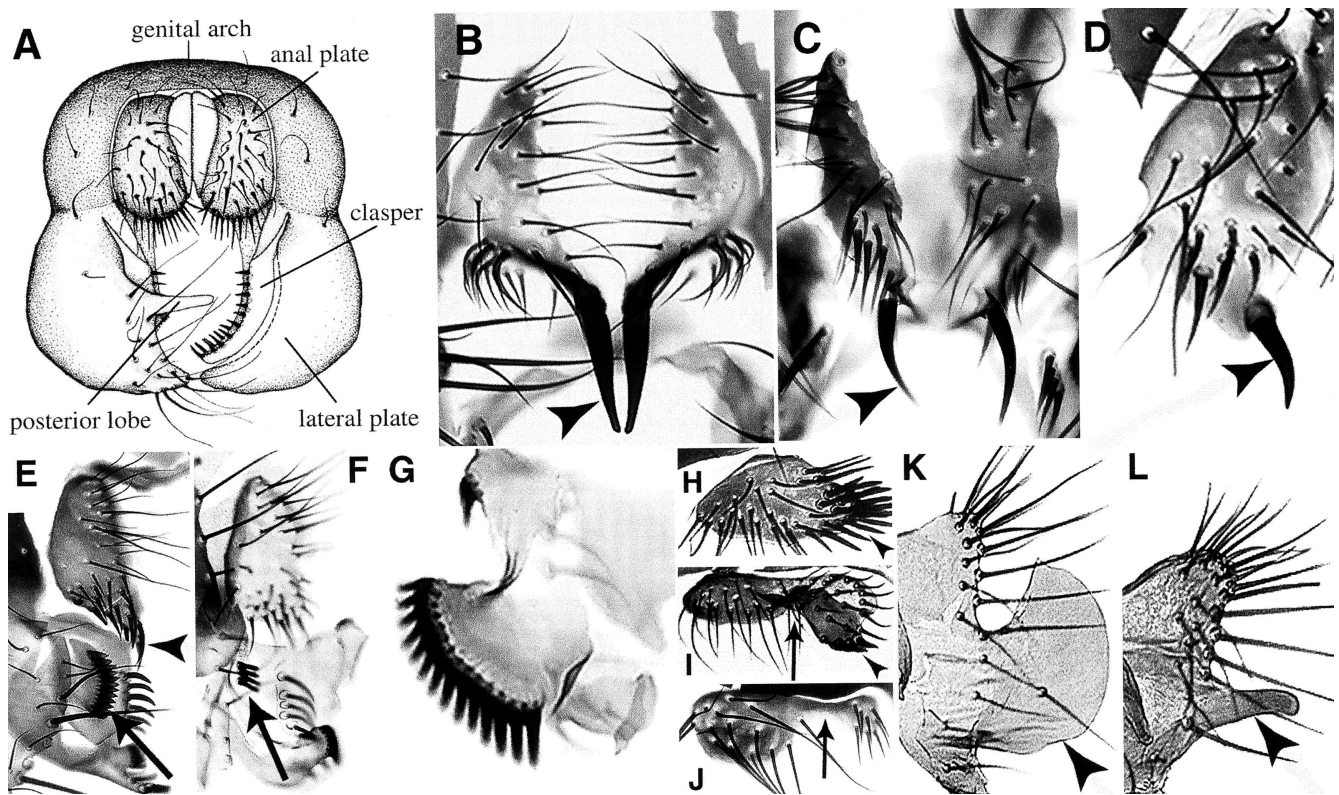


Fig. 7. External male genitalia and analia. (A) A schematic drawing of the male terminalia of *D. melanogaster*, posterior view (reproduced from True et al. 1997, with permission). (B and C) Anal plates of *D. elegans* and *D. lucipennis*, respectively (posterior views). Note the single enlarged bristle on the ventral-posterior tip of each anal plate (arrowheads). (D) A similar bristle is seen on the anal plate of *D. mimetica* (arrowhead). (E and F) Claspers of *D. suzukii* and *D. biarmipes*, respectively. Note the difference in the number of clasper bristles (arrows). In *D. suzukii*, also note the thickened bristles on the ventral surface of the anal plate (arrowhead). (G) The clasper of *D. takahashii*. (H, I, and J) The anal plates of *D. mauritiana*, *D. orena*, and *D. ficusphila*, respectively. In I and J, note the constriction between the dorsal and ventral parts of the anal plate (arrows). In H, note the thickened bristles on the ventral surface of the anal plate (arrowhead), similar to *D. suzukii* (E). In I, note the cercal clasper (arrowhead). (K and L) Posterior lobes of the sibling species *D. simulans* and *D. mauritiana*, respectively (arrowheads).

CONCLUSIONS

The rapid evolution of male sexual characters is a widely acknowledged phenomenon (Andersson 1994). However, detailed phylogenetic information is essential for understanding the sequence and direction of evolutionary changes. In this report, we show that the distribution of male sexual traits in the Oriental *D. melanogaster* lineage is the result of a complex interplay of three evolutionary patterns: rapid divergence of homologous traits among closely related species, independent origin of similar morphologies in different clades, and secondary loss of male sexual characters. Our findings add to the growing number of phylogenetic studies that document both independent gain and frequent loss of sexual dimorphism in a variety of animals (Price and Birch 1996; Andersen 1997; Burns 1998; Wiens 1999; Baker and Wilkinson 2001; Eberhard 2001; reviewed in Wiens 2001). These patterns have important implications for understand-

ing such important evolutionary questions as sexual selection and developmental constraints. In particular, the frequent loss of male sexual traits suggests that female mating preferences can change equally easily, a possibility that must be taken into account in considering the coevolution between male and female reproductive strategies (Wiens 1999, 2001).

With respect to convergent or parallel evolution, it will be important to determine whether the independently evolved traits also share a similar genetic and molecular basis. Our study provides a historical framework for addressing these questions in *Drosophila* and underscores the value of phylogenetic analysis for elucidating the evolution of animal development.

Acknowledgments

We thank Sean Carroll for his unflagging support and many fruitful discussions. *Drosophila* species stocks were kindly provided by Yoshiaki Fuyama, Masahito Kimura, and the US National *Drosophila*

Species Stock Center. We are grateful to Geeta Bharathan, Sean Carroll, Nicole King, Douglas Futuyma, Gregory Wray, and two anonymous reviewers for advice and comments on the manuscript; to Martin Garment for help with electron microscopy; and to Samantha Gasson for the drawing in Fig. 7A. Supported by the Howard Hughes Medical Institute (Sean Carroll). A. K. is an HHMI fellow of the Life Sciences Research Foundation. J. R. T. was supported by an NIH postdoctoral fellowship.

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